

Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: LTZGLUE | Context: Luxembourg Public Administration — Civil Servant NLU Deployment

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Benchmark inputs are drawn predominantly from RTL news articles, the Luxembourgish Online Dictionary, Wikipedia/Wikidata, and a translated English voice-assistant corpus [Q12, Q16, Q24, Q30, Q34, Q46, Q51]. The authors explicitly warn that 'most data sources reflect formal writing or institutional usage and therefore do not fully represent informal and multilingual contexts' [Q112], and that LTZ data 'often originate from a limited set of public domains' [Q119]. The deployment, however, expects orthographically variable, code-switched (Luxembourgish/French/German) citizen correspondence [elicitation Q2; regional context languages.code_switching_patterns]. Only the human-annotated NER sub-corpus from RTL user comments provides partial coverage of informal and code-mixed writing [Q41]. The intent detection corpus uses a voice-assistant register that has no LTZ reference, so terminology could not be systematically verified [Q55, Q56]. Given that ~47% of Luxembourg's workforce are frontaliers (mostly French-speaking) [WEB-5, WEB-2] and the language law requires the administration to respond in the citizen's chosen language [WEB-14], the benchmark's content distribution is materially narrower than the deployment input distribution. IC is HIGH priority and the documented gaps are substantial.

ID	Evidence
Q2	"Our tasks include common natural language processing tasks in binary and multi-class classification settings, including named entity recognition, topic classification, and intent classification." (p.1)
Q12	"Unless stated otherwise, the textual data used across most tasks stems from two main sources: (i)" (p.2)
Q16	"To construct this dataset, we use RTL news articles. We keep only documents from the twenty most frequent categories. We then filter articles by body length and title length, remove exact duplicate titles, randomly shuffle the remaining instances, and retain a fixed subset of 30k examples." (p.3)
Q24	"We use articles from RTL, randomly selected from the commentary and letter to the editor sections." (p.3)
Q30	"The sentences are derived from the Luxembourgish Online Dictionary (LOD) and are manipulated using the tags available in the dataset." (p.3)
Q34	"The JUDGEWEL dataset (Plum et al., 2026) introduces an automatically constructed corpus for named entity recognition (NER) in LTZ, derived from Wikipedia and Wikidata." (p.4)
Q41	"It covers a wider range of text types and registers, including informal and code-mixed writing, and focuses on four primary entity categories (PER, ORG, LOC, GPE)." (p.4)
Q46	"To construct the news topic classification (TC) dataset, we collected news articles from RTL, which provides content pre-assigned to editorial categories." (p.4)
Q51	"We constructed a new LTZ dataset for intent detection (ID) by translating the English xSID test and validation datasets (van der Goot et al., 2021)." (p.4)
Q55	"The main challenge in translating the English dataset stems from its register. The source segments consist of user commands for a voice-controlled AI assistant, representing a specialised spoken register for which there is no equivalent reference corpus in LTZ. This register is marked by domain-specific terminology and collocations (e.g., set an alarm, set a reminder, add to playlist), as well as non-standard spelling (e.g., all lowercase, missing punctuation)." (p.5)
Q56	"Due to the lack of LTZ references in this register, it was not possible to systematically verify the translated terminology." (p.5)
Q112	"Coverage across domains, registers, and demographic varieties may also be limited. LTZ displays substantial orthographic and sociolinguistic variation, yet most data sources reflect formal writing or institutional usage and therefore do not fully represent informal and multilingual contexts." (p.9)
Q119	"LTZ is a small language community, and linguistic data often originate from a limited set of public domains." (p.10)

WEB-2	OECD 2025: 50% of frontaliers from France, remainder split between Belgium and Germany — supports differentiated administrative-query profile (https://www.oecd.org/en/publications/2025/04/oecd-economic-surveys-luxembourg-2025_3eb782b5/full-report/reviving-productivity-growth_4509ab88.html)
WEB-5	~231,290 frontaliers (~47% of employment) — citizen-input distribution will be heavily multilingual (https://en.paperjam.lu/article/growth-in-number-of-cross-bord)
WEB-14	Language law requires response in citizen's chosen language — implies language-of-input output dimension absent from benchmark (https://data.legilux.public.lu/filestore/eli/etat/leg/loi/1984/02/24/n1/jo/fr/html/eli-etat-leg-loi-1984-02-24-n1-jo-fr.html)

Strengths

- Lothritz et al. NER sub-corpus from RTL online news comments provides partial informal and code-mixed writing coverage [Q40, Q41]
- Authors transparently document content limitations, register narrowness, and demographic-coverage caveats [Q110, Q112, Q119, Q121]
- Pre-training corpus for LTZ-E1 includes broader registers (user comments, webchat, transcribed podcasts and political speeches) [Q74], which may benefit downstream model representations even if task evaluation is narrower

ID	Evidence
Q40	"The NER dataset introduced by Lothritz et al. (2022), by contrast, is a fully human-annotated corpus derived from RTL online news comments." (p.4)
Q41	"It covers a wider range of text types and registers, including informal and code-mixed writing, and focuses on four primary entity categories (PER, ORG, LOC, GPE)." (p.4)
Q74	"The pre-training set is compiled from a variety of sources of LTZ. A large portion of the data stems from RTL (see Section 3), including news articles (News), transcribed radio interviews (Radio), and user comments (Comments). We also include transcribed podcasts (Podcasts) and transcribed political speeches and debates from the Chambre des Députés (Chamber). In addition, we use 1M sentences from the web crawl of the Leipzig Collection (Web, this excludes RTL), text crawled from LTZ chat rooms (Webchat), a Wikipedia crawl from October 2023 (Wikipedia), and finally, example sentences from the LOD retrieved in March 2024." (p.6)
Q110	"While LTZGLUE provides the first systematic benchmark for LTZ NLU, the dataset remains constrained by the availability and scope of existing resources. Several tasks rely on relatively small or domain-specific corpora, which limits the ecological validity of the results and restricts the range of linguistic phenomena covered." (p.9)
Q112	"Coverage across domains, registers, and demographic varieties may also be limited. LTZ displays substantial orthographic and sociolinguistic variation, yet most data sources reflect formal writing or institutional usage and therefore do not fully represent informal and multilingual contexts." (p.9)
Q119	"LTZ is a small language community, and linguistic data often originate from a limited set of public domains." (p.10)
Q121	"We therefore caution against using benchmark performance as evidence of cultural or demographic coverage." (p.10)

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — citizen correspondence requires handling of orthographically variable Luxembourgish, French/German code-switching, frontier-specific terminology, and Luxembourg-specific institutional vocabulary [elicitation Q2; regional context languages.code_switching_patterns, cultural_norms_notes]. The benchmark inputs are dominated by RTL news and dictionary content [Q12, Q16, Q46], which do not encode this distribution.

IC-2: Partially. Sources are Luxembourgish in origin so cultural framing is locally grounded, but the institutional/journalistic register diverges from citizen-government correspondence [Q112]. The voice-assistant intent corpus is translated from English without LTZ register reference [Q55, Q56], introducing alien register material.

IC-3: Yes — the xSID-derived intent corpus (alarm/reminder/weather) [Q51, Q55] is Western voice-assistant content with no administrative analogue in Luxembourg; its inputs encode construct-irrelevant register. Authors note pre-training data overlap with LLMs may also inflate performance [Q111].

IC-4: INSUFFICIENT DOCUMENTATION — no Luxembourg civil-administration domain experts appear to have reviewed task inputs for representativeness; the elicitation flags this as needing real-time civil-servant feedback for progressive alignment [elicitation Q3].

IC-5: Documented content-validity issues: (a) formal/institutional register dominance [Q112], (b) absence of systematic code-switching coverage outside NER [Q41], (c) voice-assistant register import without LTZ reference [Q55, Q56], (d) no representation of frontier-specific administrative content [WEB-5].

ID	Evidence
Q2	“Our tasks include common natural language processing tasks in binary and multi-class classification settings, including named entity recognition, topic classification, and intent classification.” (p.1)
Q3	“We evaluate various pre-trained language models for LTZ to present an overview of the current capabilities of these models on the LTZ language.” (p.1)
Q12	“Unless stated otherwise, the textual data used across most tasks stems from two main sources: (i)” (p.2)
Q16	“To construct this dataset, we use RTL news articles. We keep only documents from the twenty most frequent categories. We then filter articles by body length and title length, remove exact duplicate titles, randomly shuffle the remaining instances, and retain a fixed subset of 30k examples.” (p.3)
Q41	“It covers a wider range of text types and registers, including informal and code-mixed writing, and focuses on four primary entity categories (PER, ORG, LOC, GPE).” (p.4)
Q46	“To construct the news topic classification (TC) dataset, we collected news articles from RTL, which provides content pre-assigned to editorial categories.” (p.4)
Q51	“We constructed a new LTZ dataset for intent detection (ID) by translating the English xSID test and validation datasets (van der Goot et al., 2021).” (p.4)
Q55	“The main challenge in translating the English dataset stems from its register. The source segments consist of user commands for a voice-controlled AI assistant, representing a specialised spoken register for which there is no equivalent reference corpus in LTZ. This register is marked by domain-specific terminology and collocations (e.g., set an alarm, set a reminder, add to playlist), as well as non-standard spelling (e.g., all lowercase, missing punctuation).” (p.5)
Q56	“Due to the lack of LTZ references in this register, it was not possible to systematically verify the translated terminology.” (p.5)
Q111	“In addition, some of the data sources used in this benchmark may already be included, in whole or in part, in the pre-training corpora of the large language models evaluated in this work. While the exact composition of proprietary pre-training datasets is typically not fully disclosed, this potential overlap cannot be entirely ruled out and may inflate performance estimates.” (p.9)
Q112	“Coverage across domains, registers, and demographic varieties may also be limited. LTZ displays substantial orthographic and sociolinguistic variation, yet most data sources reflect formal writing or institutional usage and therefore do not fully represent informal and multilingual contexts.” (p.9)
WEB-5	~231,290 frontaliers (~47% of employment) — citizen-input distribution will be heavily multilingual (https://en.paperjam.lu/article/growth-in-number-of-cross-bord)
WEB-14	Language law requires response in citizen's chosen language — implies language-of-input output dimension absent from benchmark (https://data.legilux.public.lu/filestore/eli/etat/leg/loi/1984/02/24/n1/jo/fr/html/eli-etat-leg-loi-1984-02-24-n1-jo-fr-html.html)

Information Gaps

- No quantification of code-switching frequency in any task split
- No measure of orthographic-variation coverage in task inputs
- Whether RTL user-comment NER sub-corpus contains enough French/German switching to represent deployment distribution

Requires Expert Verification

- Sampling and qualitative review of LTZGLUE task inputs by civil-administration linguists to confirm representativeness for citizen-correspondence content
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Remediation

Gap 1: Benchmark inputs are dominated by formal/institutional registers and filter out non-Luxembourgish text; code-switching and orthographic variation are under-represented

Recommendation: Construct an internal evaluation corpus from anonymized real citizen correspondence (with CNPD-compliant DPIA) that explicitly preserves French/German code-switching, frontalier terminology, and orthographic variation. Use it as an out-of-distribution stress test alongside LTZGLUE.